

Set 11: Product Category Analysis

DSA – Data Handling and Visualization | Lab Experiment

R.No	Reg. No	Name
14	192524098	Divyasree Muthukumar

Aim

To analyse sales across product categories using a funnel chart, a labelled table and a pie chart, and to combine them in an interactive Tableau dashboard.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Category	Sales (in \$)
Electronics	50,000
Clothing	35,000
Appliances	40,000

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
products <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
write.csv(products, "Product_Categories.csv", row.names = FALSE) # for Tableau
pal <- c(Electronics = "#0072B2", Appliances = "#009E73", Clothing = "#E69F00")
```

Q1. Funnel Chart – Sales by Product Category

Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

R Program:

```
# Q1. Funnel chart - categories ranked by sales (largest stage at the top)
funnel <- products[order(-products$Sales), ]
funnel$Stage <- paste0("Stage ", seq_len(nrow(funnel)), ": ", funnel$Category)
funnel$Pct_of_Top <- round(100 * funnel$Sales / max(funnel$Sales), 1)
funnel$Drop_from_Prev <- c(NA, round(100 * (1 - funnel$Sales[-1] /
  funnel$Sales[-nrow(funnel)]), 1))
print(funnel, row.names = FALSE)

funnel$Stage <- factor(funnel$Stage, levels = rev(funnel$Stage))
p_fun <- ggplot(funnel, aes(x = Stage)) +
  geom_col(aes(y = Sales / 2, fill = Category), width = 0.85) +
  geom_col(aes(y = -Sales / 2, fill = Category), width = 0.85) +
  geom_text(aes(y = 0, label = paste0(Category, "\n",
    format(Sales, big.mark = ","), " (",
    Pct_of_Top, "%)")),
    colour = "white", fontface = "bold", size = 4) +
  coord_flip() +
  scale_fill_manual(values = pal) +
  scale_y_continuous(labels = NULL) +
  labs(title = "Sales Funnel by Product Category",
    subtitle = "Highest to lowest sales; % = share of top stage",
    x = "Funnel Stage", y = NULL) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"), legend.position = "none",
    panel.grid = element_blank())
```

```
print(p_fun)
ggsave("s11_q1_funnel.png", p_fun, width = 7.5, height = 4.3, dpi = 150, bg = "white")
```

Console Output:

Category	Sales	Stage	Pct_of_Top	Drop_from_Prev
Electronics	50000	Stage 1: Electronics	100	NA
Appliances	40000	Stage 2: Appliances	80	20.0
Clothing	35000	Stage 3: Clothing	70	12.5

Output:

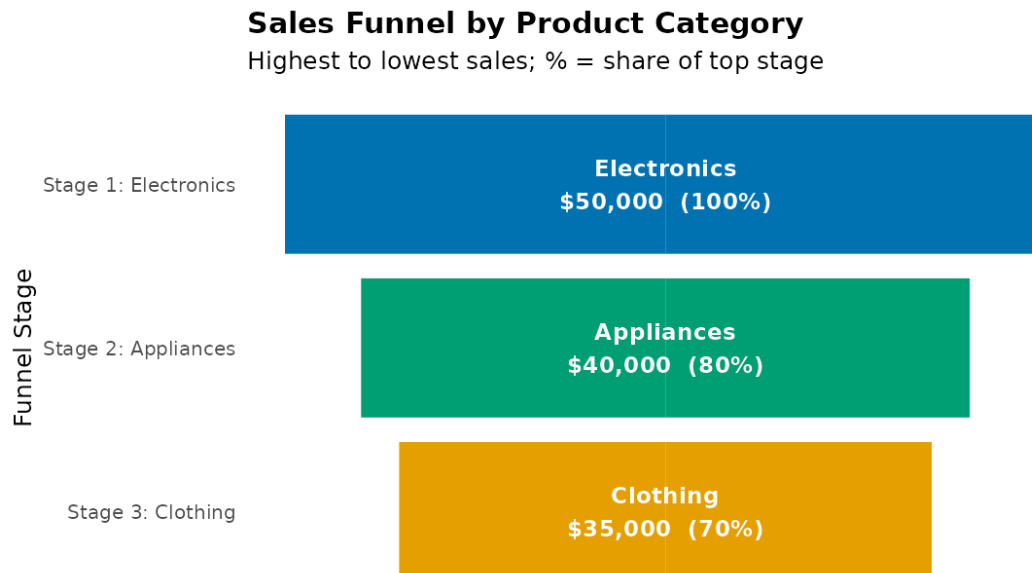


Figure 1: Sales funnel with categories ranked by sales

Inference:

The dataset has one sales value per category and no stage-wise data (such as visits → carts → purchases), so the funnel ranks the categories from highest to lowest sales. Each stage is labelled with its share of the top stage.

Electronics is the widest stage (\$50,000, 100%). Appliances follows at 80% (a 20% drop) and Clothing at 70% (a further 12.5% drop). Electronics leads sales, and Clothing has the most room to grow.

Q2. Table – Sales Data for Each Product Category

Build a table to display the sales data for each product category. Label the table elements.

R Program:

```
# Q2. Labelled table of sales by category
tbl <- products
tbl$Share <- paste0(round(100 * tbl$Sales / sum(tbl$Sales), 1), "%")
tbl$Sales <- paste0("$", format(tbl$Sales, big.mark = ","))
tbl <- rbind(tbl, data.frame(Category = "Total",
                             Sales = paste0("$", format(sum(products$Sales), big.mark = ",")),
                             Share = "100%"))
names(tbl) <- c("Product Category", "Sales (in $)", "Share of Total Sales")
print(tbl, row.names = FALSE)
tg <- tableGrob(tbl, rows = NULL, theme = ttheme_default(
  core = list(fg_params = list(fontsize = 11)),
  colhead = list(fg_params = list(fontsize = 12, fontface = "bold", col = "white"),
    bg_params = list(fill = "#1F3864"))))
title <- grid::textGrob("Sales Data by Product Category",
  gp = grid::gpar(fontsize = 14, fontface = "bold"))
png("s11_q2_table.png", width = 900, height = 380, res = 150)
grid.arrange(title, tg, ncol = 1, heights = c(0.25, 1))
invisible(dev.off())
```

Console Output:

Product Category	Sales (in \$)	Share of Total Sales
Electronics	\$50,000	40%
Clothing	\$35,000	28%

Appliances	\$40,000	32%
Total	\$125,000	100%

Output:

Sales Data by Product Category

Product Category	Sales (In \$)	Share of Total Sales
Electronics	\$50,000	40%
Clothing	\$35,000	28%
Appliances	\$40,000	32%
Total	\$125,000	100%

Figure 2: Labelled sales table with share of total

Inference:

Total sales are \$125,000. Electronics contributes 40%, Appliances 32% and Clothing 28%.

Q3. Tableau Dashboard – Pie Chart, Funnel Chart and Table

Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

R Program:

```
# Q3. Dashboard layout preview (Tableau dashboard built from Product_Categories.csv)
library(patchwork)
dash <- (p_pie | p_fun) / wrap_elements(full = tg) +
  plot_layout(heights = c(1.4, 0.6)) +
  plot_annotation(title = "Product Category Dashboard",
    theme = theme(plot.title = element_text(face = "bold", size = 16)))
ggsave("s11_dashboard.png", dash, width = 12, height = 7, dpi = 130, bg = "white")
```

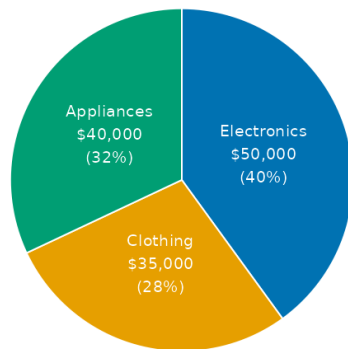
Procedure in Tableau:

- 1 Open Tableau → **Connect** → **Text file** → **Product_Categories.csv**.
- 2 **Sheet 1 – "Sales Pie"**: set the Marks type to **Pie**. Drag Category to Color and SUM(Sales) to Angle. Drag Category, SUM(Sales) and a **Percent of Total** quick table calculation to Label.
- 3 **Sheet 2 – "Sales Funnel"**: create a calculated field **Negative Sales = -SUM([Sales])**. Put Category on Rows, sorted **Descending by Sales**. Put Negative Sales and SUM(Sales) side by side on Columns, use the same colour (Category) on both panes, and hide the axis headers. This gives a symmetric funnel. Add Sales and Category labels.
- 4 **Sheet 3 – "Sales Table"**: drag Category to Rows and SUM(Sales) to Text, then add a Percent of Total column. Turn on **Analysis** → **Totals** → **Show Column Grand Totals**.
- 5 From the menu choose **Dashboard** → **New Dashboard**. Set **Size** → **Automatic** (or Fixed 1200 × 800).
- 6 Drag the sheets onto the canvas: Sales Pie (top-left), Sales Funnel (top-right), Sales Table (bottom). Add a **Text** object at the top for the dashboard title.
- 7 On each chart, click the sheet's drop-down and select **Use as Filter**. Clicking a bar or slice then filters the other views.
- 8 Go to **Dashboard** → **Actions** → **Add Action** → **Highlight** (Run on: Hover) so related marks highlight across all views.
- 9 Right-click a filter → **Apply to Worksheets** → **All Using This Data Source** (Product_Categories) so one filter controls every view.
- 10 Save as a packaged workbook (**File** → **Save As** → **.twbx**) or publish to Tableau Public.

Dashboard Layout:

Product Category Dashboard

Sales Share by Product Category



Category
Appliances
Clothing
Electronics

Funnel Stage

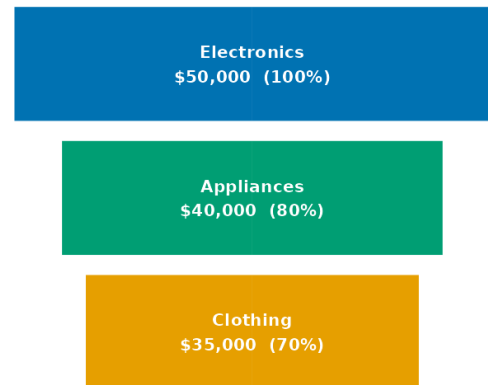
Stage 1: Electronics

Stage 2: Appliances

Stage 3: Clothing

Sales Funnel by Product Category

Highest to lowest sales; % = share of top stage



Product Category	Sales (in \$)	Share of Total Sales
Electronics	\$50,000	40%
Clothing	\$35,000	28%
Appliances	\$40,000	32%
Total	\$125,000	100%

Reference layout of the dashboard (rendered in R). The interactive version is built in Tableau using the procedure above.

Inference:

Selecting a category slice in the pie highlights its funnel stage and its table row, so its absolute sales, rank and share can be read together.

Q4. Pie Chart – Distribution of Sales across Categories

Create a pie chart to represent the distribution of sales across product categories. Include labels.

R Program:

```
# Q4. Pie chart - share of sales by category, with labels
products$Percent <- round(100 * products$Sales / sum(products$Sales), 1)
print(products, row.names = FALSE)
p_pie <- ggplot(products, aes(x = "", y = Sales, fill = Category)) +
  geom_col(width = 1, colour = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = paste0(Category, "\n$", format(Sales, big.mark = ","),
    "\n(", Percent, "%)")),
    position = position_stack(vjust = 0.5), colour = "white", size = 3.8) +
  scale_fill_manual(values = pal) +
  labs(title = "Sales Share by Product Category", fill = "Category") +
  theme_void(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5))
print(p_pie)
ggsave("s11_q4_pie.png", p_pie, width = 6, height = 4.8, dpi = 150, bg = "white")
```

Console Output:

```
Category Sales Percent
Electronics 50000      40
Clothing 35000      28
Appliances 40000      32
```

Output:

Sales Share by Product Category

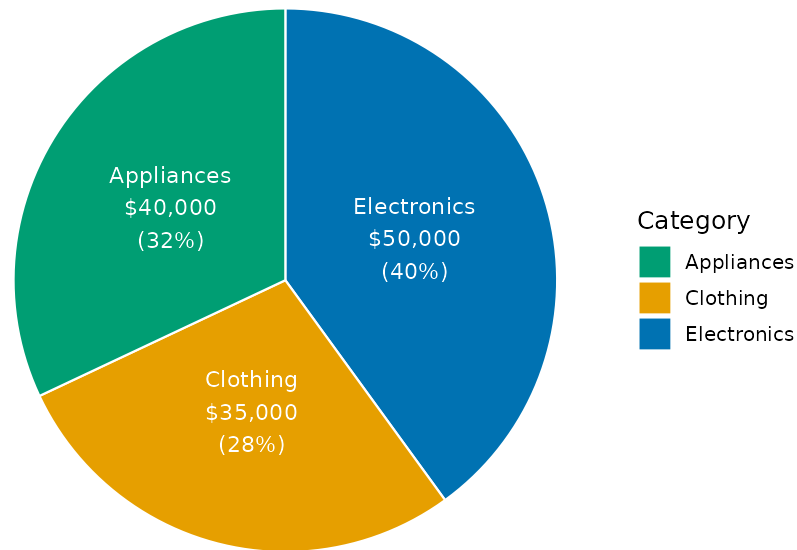


Figure 3: Share of sales by product category

Inference:

Electronics has the largest share of sales (40%, \$50,000), followed by Appliances (32%, \$40,000) and Clothing (28%, \$35,000). No category is dominant, but Electronics is the main revenue driver.

Result

The product category sales were visualised with a funnel chart, a labelled table and a pie chart in R, and combined into an interactive Tableau dashboard. Electronics leads with 40% of the \$125,000 total, followed by Appliances (32%) and Clothing (28%).